

Advanced Data Engineering with Google Cloud, Python & SQL

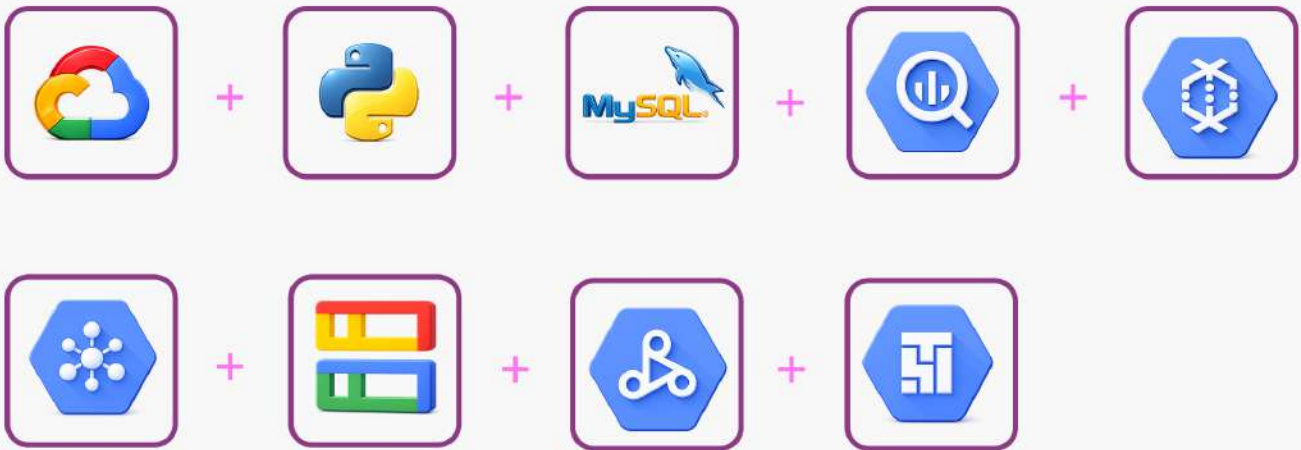


Course Curriculum

Introduction to Data Engineering

This Industry-Oriented Program Is Designed For Professionals And Associates Who Want To Master Modern Data Engineering Using Google Cloud Platform, Python, And SQL. The Curriculum Focuses On Real-World Batch And Streaming Pipelines, Data Warehousing, Orchestration, And Production Best Practices.

Tools & Technologies Covered



Module-1: Modern Data Engineering Foundations & Data Warehousing

- Role And Responsibilities Of A Modern Data Engineer
- Data Lifecycle: Ingestion, Processing, Storage, Analytics
- Common Data Architecture Patterns (Lambda, Kappa)
- Batch Vs Streaming Processing Systems
- OLTP Vs OLAP Systems And Use Cases
- Google Cloud Platform Overview
- GCP Project Setup And Billing Basics
- IAM Fundamentals And Service Accounts

- Cloud Storage Buckets And Lifecycle Management

Module-2: Python for Data Engineering

- Python Fundamentals And Control Structures
- Functions And Modular Programming
- Object-Oriented Programming For Pipelines
- Handling CSV, JSON, And Parquet Files
- Local And Cloud Storage Integration
- Logging Standards And Error Handling
- Building Reusable Ingestion Frameworks

Module-3: ETL Development & API Integration

- ETL Vs ELT Architectures
- Schema Validation And Data Quality Checks
- Handling Missing, Late, And Corrupt Data
- Working With REST APIs And Authentication
- Transformations Using Pandas
- Introduction To PySpark
- Automated ETL Workflows

Module-4: SQL for Data Engineering

- Relational Database Fundamentals
- Joins, Subqueries, And CTEs
- Aggregations And Grouping Techniques
- Query Performance Optimization Basics
- Hands-On Analytical SQL Use Cases

Module-5: Advanced SQL & Data Warehousing

- Window Functions And Analytical Queries
- Star And Snowflake Schema Modeling
- Fact And Dimension Table Design
- Slowly Changing Dimensions (SCD)
- Implementing SCD Type 2
- Enterprise Data Warehouse Best Practices

Module-6: BigQuery Deep Dive

- BigQuery Architecture And Internals
- Loading Data From Cloud Storage
- Partitioning And Clustering Strategies
- Query Execution Plan Analysis
- Cost Optimization Techniques
- Monitoring And Performance Tuning

Module-7: Streaming with Pub/Sub & Dataflow

- Streaming Data Fundamentals
- Pub/Sub Topics And Subscriptions
- Message Ordering And Delivery Semantics
- Apache Beam Programming Model
- Windowing, Triggers, And Watermarks
- Dataflow Pipeline Design
- Streaming Data Ingestion Into BigQuery

Module-8: Orchestration & Production Best Practices

- Workflow Orchestration Concepts
- Apache Airflow Fundamentals
- Cloud Composer Architecture
- Building And Scheduling DAGs
- Monitoring, Logging, And Alerting
- Security And IAM Best Practices
- CI/CD For Data Pipelines

Module-9: Real-Time Project & Certification Preparation

- Understanding Real-World Data Engineering Scenarios

- Designing End-To-End Data Pipeline Architecture
- Ingesting Data From APIs And File
- Processing Data Using Python And Dataflow
- Loading Curated Data Into BigQuery
- Orchestrating Workflows Using Airflow
- Performance And Cost Optimization
- Documentation And Architecture Diagrams
- Interview And Certification Discussion

Course Durations

Total Duration: 2 Months